The extremal maximum-likelihood degree bound

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Abstract

We prove that the extremal maximum-likelihood degree of a graph is at most its Gaussian maximum-likelihood degree. The proof deforms each generic complex extremal critical point into a nonsingular Gaussian critical point. A second-order determinant expansion resolves the degeneracy in the added diagonal parameters.

Task 1: statement and likelihood coordinates

The target is Conjecture 4.12 of [1], page 19. Degrees count complex critical points for generic data, on the domain where the relevant determinant is nonzero.

Theorem 1. *For every connected finite simple graph G,*

$$\mathrm{eMLD}(G) \le \mathrm{MLD}(G).$$

The same bound holds for disconnected graphs under either the strict completion definition in the source or its componentwise extension.

Let G have vertices $1, \dots, n$ and edge set E. We first take $n \ge 2$ and G connected. Give its edges complex weights w_e and set

$$L(w) = \sum_{e=\{i,j\}\in E} w_e (e_i - e_j)(e_i - e_j)^{\mathsf{T}}, \quad A(w) = L(w)_{\widehat{n},\widehat{n}}, \quad \tau(w) = \det A(w).$$

Here a hat means that the row and column are deleted. The map from weights to A is injective: a Laplacian with zero row sums is determined by this principal submatrix, and its off-diagonal entries determine the weights. On $\tau \neq 0$, the extremal score equations are

$$\partial_{w_e} \log \tau(w) = v_e \qquad (e \in E). \tag{1}$$

Indeed, the pseudo-determinant of L is $n\tau$, and for a hollow variogram Γ,

$$\tfrac{1}{2} \operatorname{tr}(L\Gamma) = -\sum_{\{i,j\}\in E} w_{ij}\Gamma_{ij}.$$

Thus (1) is exactly the critical system for the source's surrogate likelihood, with $v_{ij} = \Gamma_{ij}$; see [1, Section 4, equation (16)]. It also describes the spanning-tree concentration model in [2].

Every symmetric Gaussian concentration matrix with this graph's off-diagonal zero pattern has unique coordinates

$$K = L(w) + \operatorname{diag}(s_1, \dots, s_n), \qquad F(w, s) = \det K.$$

For a symmetric sample matrix S, set $u_i = S_{ii}$ and $v_{ij} = S_{ii} + S_{jj} - 2S_{ij}$ on edges. These are invertible linear coordinates on the specified sample entries, and

$$\operatorname{tr}(KS) = \sum_i s_i u_i + \sum_e w_e v_e.$$

Consequently the Gaussian score equations are

$$\partial_{w_e} \log F = v_e, \qquad \partial_{s_i} \log F = u_i, \qquad F \neq 0. \tag{2}$$

For completeness, generic fibers of both score maps are finite and nonsingular. At positive real edge weights, A is positive definite. The Hessian of $\log\det A$ in a nonzero weight direction $\dot{A}$ is

$$-\operatorname{tr}(A^{-1}\dot{A}A^{-1}\dot{A}) < 0,$$

because the coordinate map is injective. Thus the score map has a nonzero Jacobian determinant. It is a dominant rational map between spaces of the same dimension, generically finite and unramified over $\mathbb{C}$. The identical argument at a positive definite K applies to (2). We may therefore choose v such that (1) has exactly $r = \mathrm{eMLD}(G)$ distinct solutions, all with $\tau \neq 0$ and nonsingular Hessian $\nabla_w^2 \log \tau$.

The determinant expansion

Fix any w with $\tau(w) \neq 0$ and put $C = A^{-1}$. Extend C to an n-by-n matrix $\widetilde{C}$ by giving it a zero last row and column. Define

$$R_{ij} = \widetilde{C}_{ii} + \widetilde{C}_{jj} - 2\widetilde{C}_{ij}, \qquad Z = \sum_{i=1}^{n} z_i. \tag{3}$$

All these functions are holomorphic near the chosen w.

Lemma 1. *As a polynomial identity in the parameter t, locally on $\tau \neq 0$,*

$$F(w, tz) = t\tau Z + t^2 \tau \sum_{i<j} R_{ij} z_i z_j + O(t^3). \tag{4}$$

Here the remainder is divisible by t^3.

Proof. Write $\widehat{z} = (z_1, \ldots, z_{n-1})^{\mathsf{T}}$ and $D = \operatorname{diag}(\widehat{z})$. The zero row sums give

$$L = \begin{pmatrix} A & -A\mathbf{1} \\ -\mathbf{1}^{\mathsf{T}}A & \mathbf{1}^{\mathsf{T}}A\mathbf{1} \end{pmatrix}.$$

The Schur complement of $A + tD$ in $L + t\operatorname{diag}(z)$ equals

$$tZ - t^2 \widehat{z}^{\mathsf{T}} C \widehat{z} + O(t^3),$$

while

$$\det(A + tD) = \tau \left(1 + t \sum_{i<n} C_{ii} z_i + O(t^2)\right).$$

Multiplying proves (4), since

$$Z \sum_{i<n} C_{ii} z_i - \widehat{z}^{\mathsf{T}} C \widehat{z} = \sum_{i<j} R_{ij} z_i z_j.$$

□

Lifting every extremal critical point

Since $F(w,0)=0$, the quotient

$$P(t,w,z)=\frac{F(w,tz)}{t}$$

is a polynomial, including at $t=0$. Consider the following $|E|+n$ equations in the same number of unknowns (w,z):

$$H_e=\partial_{w_e}P-v_eP=0 \qquad (e\in E), \tag{5}$$

$$H_0=\partial_{z_n}P-P=0, \tag{6}$$

$$H_i=\frac{\partial_{z_i}P-\partial_{z_n}P}{t}=0 \qquad (1\le i<n). \tag{7}$$

The last expressions are polynomials because $P(0,w,z)=\tau(w)Z$, whose z_i derivatives all equal τ. When $t\neq 0$ and $P\neq 0$, these equations are equivalent to (2) for

$$s_i=tz_i, \qquad u_i=1/t \quad (1\le i\le n), \qquad v_e \text{ as fixed above.} \tag{8}$$

Let w^* be any of the r chosen extremal critical points. At $t=0$, Lemma 1 turns (5)–(7) into

$$\begin{aligned}
H_e &= Z(\partial_{w_e}\tau-v_e\tau),\\
H_0 &= \tau(1-Z),\\
H_i &= \tau\big((Rz)_i-(Rz)_n\big)=\tau\left(C_{ii}Z-2(C\widehat{z})_i\right).
\end{aligned}$$

Thus they have the unique solution in z given by

$$Z=1, \qquad \widehat{z}^*=\tfrac{1}{2}A\,(C_{11},\dots,C_{n-1,n-1})^{\mathsf{T}}, \qquad z_n^*=1-\sum_{i<n}z_i^*. \tag{9}$$

Use coordinates $(w,\widehat{z},Z)$ and order the equations as $(H_e)_{e\in E},(H_i)_{i<n},H_0$. Their Jacobian at $(t,w,z)=(0,w^*,z^*)$ is block lower triangular. Its upper diagonal block is

$$\tau\,\nabla_w^2\log\tau,$$

and its lower diagonal block is

$$\begin{pmatrix} -2\tau C & \tau\,(C_{11},\dots,C_{n-1,n-1})^{\mathsf{T}} \\ 0 & -\tau \end{pmatrix}. \tag{10}$$

Both are nonsingular. In the upper right block, derivatives with respect to z vanish because (1) holds. This verifies the triangular claim without any assumption on the signs of the complex weights or on the entries of z^*.

The holomorphic implicit-function theorem now gives a solution $(w(t),z(t))$ near each (w^*,z^*) for all sufficiently small t. Its Jacobian stays nonsingular, and $P(t,w(t),z(t))\to\tau(w^*)\neq 0$. Because the r limiting points are distinct, their neighborhoods may be chosen disjoint. For a common sufficiently small nonzero t, we obtain r distinct nonsingular solutions of (2). The changes of variables and equations used above are invertible for this fixed $t\neq 0$.

The special data in (8) need not themselves be generic. Each of these r nonsingular Gaussian solutions persists under a small arbitrary perturbation of all data coordinates (u,v), again by the implicit-function theorem. The perturbation can be chosen in the Zariski-open set where the

number of solutions equals $\mathrm{MLD}(G)$. It follows that $r \leq \mathrm{MLD}(G)$, proving the bound for connected G with $n \geq 2$.

For a single vertex both degrees are one. If G has two or more connected components, every graph-supported Laplacian has at least two independent kernel vectors, the component indicators. Thus it cannot have rank $n-1$. The source's strict variogram model, and its regular complex completion domain, are then empty, giving degree zero and a trivial inequality. Alternatively, if disconnected extremal models are defined componentwise, their degrees multiply, just as Gaussian degrees do. Multiplication of the connected inequalities proves the bound for that extension as well. This completes Theorem 1.

Verification and proposed follow-up tasks

The exact verifier checks the determinant coefficients and the two Jacobian blocks on all connected labeled graphs with two through four vertices, and additional graphs through seven vertices. These finite calculations supplement the all-graph proof above.

Task 2 (proposed). Classify connected graphs for which $\mathrm{eMLD}(G) = \mathrm{MLD}(G)$, beyond the known chordal case.

Task 3 (proposed). Describe and count the Gaussian critical points that do not converge to the extremal branches constructed here, in order to express $\mathrm{MLD}(G) - \mathrm{eMLD}(G)$ geometrically.

Task 4 (proposed). Determine whether extremal ML degree is monotone under connected induced subgraphs, as suggested by the spanning-tree-model questions in [2].

Task 5 (proposed). Extend the deformation to concentration spaces with a fixed kernel of dimension greater than one, identifying conditions that make the analogue of (10) nonsingular.

References

[1] C. Améndola, J. I. Coons, A. Grosdos, and F. Röttger, *Algebraic statistics of Hüsler–Reiss graphical models in multivariate extremes*, arXiv:2603.02191v1 (2026), especially Conjecture 4.12. `https://arxiv.org/abs/2603.02191`.

[2] S. Cox, P. Misra, and P. Semnani, *Homaloidal polynomials and Gaussian models of maximum likelihood degree one*, Algebraic Statistics **15** (2024), 167–198. `https://doi.org/10.2140/astat.2024.15.167`. Final author version: `https://arxiv.org/abs/2402.06090v3`.

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